

Also Remember...

■ Warranties from manufacturers are usually valid for one year. However, some manufacturers such as Microtek and Samsung offer a 3-year warranty on their products.

■ Integrated speakers, USB ports and a user friendly On Screen Display are other features that you might want to consider.

■ Models that come with flat picture tubes are generally more expensive.

Differences between CRT and LCD Monitors

Technology gets cheaper and better as time goes by. This is evident from the fact that faster and better processors are now available for much less than a few years ago. The same is the case with display technology. Some years back, LCD screens cost more than Rs 50,000, but now they come for much less—and with many more features. CRT monitors too have become cheaper; a 15-inch monitor can be had for as little as Rs 5,500 and a 17-inch for approximately Rs 11,000.

The technologies used to manufacture CRT and LCD are completely different. CRT monitors use a cathode ray tube which houses in it the electron gun at one end and a tough glass coated with phosphor at the other. The electron gun fires electrons towards the phosphor screen and the magnetic coil precisely deflects them from their path so that they hit the correct part of the screen. The screen is made up of phosphor dots which are placed in triad formation (in a colour monitor).

There are three electron guns; one each for red, green and blue. Each gun fires the electron on the corresponding dot. The screen is refreshed according to the refresh rate selected; if it's 75 Hz, the screen will be drawn 75 times. Due to the width of the tube inside the monitor, CRT monitors are bulkier than their LCD counterparts.

LCD displays, on the other hand, are made up of liquid crystals. The technology used is more advanced than the one in CRT displays and calls for lots of precision. To light up the pixel, LCD uses a grid to supply the charge. Two glass layers are used as a substrate. One of the substrates is used as the row, and the other as the column, and are made of a transparent conductive material. This material is mostly Indium-Tin oxide.

These rows and columns are connected to

the on-board electronics, which control the charge sent to the rows and columns. The liquid crystal material is sandwiched between the two glass substrates and the polarising material is added on the outer side of each substrate. To make the pixel work (or turn it on), a charge is sent by the integrated circuit to the correct column and ground is activated on the row of that pixel. The row and column meet each other at the pixel to be turned on—and delivers the voltage to 'untwist' the liquid crystal at that pixel.

Why does LCD cost so much?

The cost of LCD displays increases dramatically even if the screen size increases by just one inch. The main reason is that many thin film transistors have to be incorporated within the additional space. Because the technology is still in its early stages and the rejection rate is very high (in large LCDs it can be as high as 40 per cent), good LCD displays sold in the market have to bear the manufacturing cost of the bad LCD displays that get rejected. The only way out of this vicious circle is further advancement in technology and more refined manufacturing techniques.

3D TFT-LCD Monitor

Advanced graphic cards have entered the world of computers, and they are capable of rendering 3D objects in real time. To complement this, some world-renowned monitor manufacturers have made 3D TFT monitors. These monitors make users feel as if images on the screen are real.

These monitors have a glass panel that is sandwiched between two LCD panels. Each LCD panel reflects half the light, allowing permeability. This effect creates the 3D sensation. Such monitors can be used in many critical areas such as medicine, fighter aircraft





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